THE MAYFIELD HANDBOOK ON TECHNICAL AND SCIENTIFIC WRITING

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In this document, some excerpts from the "Mayfield handbook on technical and scientific writing" are collected, for educational purposes. Students are strongly encouraged to explore the whole work at http://www.mhhe.com/mayfieldpub/tsw/home.htm

Section 1

Planning and Producing Documents

Effective technical documents do not just happen; they are the result of a deliberate and comprehensive design and production process. Although writers may vary some of the steps they use to create a document, effective technical and scientific writing typically follows the same general procedures.

First, establish basic criteria. Five characteristics: accuracy, clarity, conciseness, coherence, and appropriateness, apply to all effective technical and scientific writing.

Identify the specific purpose of the document by clarifying both the reasons for its creation and its specific objectives. Often, technical and scientific documents are written as answers to a specific problem, which is articulated in a problem statement. Once you have identified the document's purpose, you should be able to determine the document's general type.

Technical documents are tools designed to be used by their readers. Accordingly, define your audience--the person or persons who will be reading the document. Then determine your audience's level of expertise and their purpose in using the document. It is also important to assess the attitude of the audience toward both you and the document's subject matter.

Once you have defined the purpose, the problem and the audience, collect, create, and assemble your information. Sketch out a preliminary outline to organize it. Keeping purpose and audience in mind, sketch out graphics and tables to display your important data.

Using your outline and preliminary graphics, write a first draft, a rough working version in which you get your ideas on paper. At this point in the process, do not be overly concerned about grammar, style, or usage.

If possible, put your first draft away for a day or two. Then revise your document in stages, saving stylistic changes for the last stage. Revise for organization; then revise the content for accuracy and appropriateness. Finally, edit your paragraphs and sentences to improve their clarity, conciseness, and coherence, and to fix any problems in grammar, spelling, punctuation, mechanics, or usage.

The last major step for most technical documents is one or more reviews. You may be too familiar with your document to see such things as gaps in information and inappropriate language. In addition, you may lack certain technical or managerial knowledge necessary for the document to

achieve its purpose. For these reasons, writers of technical and scientific documents may ask peers to review their manuscripts for accuracy, clarity, coherence, and appropriateness. In many cases, a technical expert will review the document for technical content. A technical editor may review the document to ensure that it conforms to the organization's style and to correct any remaining problems. There may be legal reviews as well. Finally, a supervisor or a manager may review the document to ensure that it achieves the organization's purpose and is appropriate to the audience.

Just as most technical documents are reviewed by several persons, many documents are written collaboratively--that is, by several individuals. Collaborative writing often involves additional steps in document planning and management, drafting, and revision.

Section 1.1

Characteristics of Effective Technical Communication

Learn to recognize and cultivate the qualities of effective technical communication.

Good technical communication is accurate, clear, concise, coherent, and appropriate. In the prose of science and technology, these qualities are sometimes difficult to achieve. Not only do science and technology depend heavily on specialized concepts and terminologies, but they also make extensive use of numbers and graphics.

The following example shows how the different qualities of technical prose work together.

The flow of electrical current can induce the migration of impurities or other defects through the bulk of a solid. This process is called electromigration. In simple electromigration, the force on the defect is thought to have two components. The first component is the force created by direct interaction between the effective charge of the defect and the electric field that drives the current. The second component, called the "wind force," is the force caused by the scattering of electrons at the defect.

--J.A. Stroscio and D.M. Eigler, "Atomic and Molecular Manipulation with the Scanning Tunneling Microscope," *Science*

The preceding example is accurate in two ways. It is stylistically accurate in its precise use of language. It is technically accurate in its use of specialized terms technical terms such as *electromigration, charge, electricfield*, and *scattering*, whose meanings are based in the context of a technical discipline. Both kinds of accuracy--accuracy of phrasing and accuracy of technical concept--are of first importance in science and technology writing.

The example is also clear because it is written in simple, direct sentences. Although the technical context is the highly specialized realm of theoretical and applied nanotechnology, the sentence syntax--word order--is restrained and structurally very simple. Part of this clarity is achieved by the rhetorical device of defining a term, *electromigration*.

The example is concise in its use of a minimum of words to express the basic idea of electromigration. It is not wordy, and it does not digress from the point being made.

The example is coherent because it develops its subject matter in an easy-to-follow line of thinking. The sentences are further linked by referents such as "this process," "the first component," and "the second component."

Finally, the example is appropriate to its purpose of presenting a general description of the process of electromigration, and to its audience, educated readers of Science, who are not necessarily experts in the field of nanotechnology.

Section 1.1.1

Accuracy

Cultivate accuracy in your writing. Accuracy, which is the careful conforming to truth or fact, hasthree main aspects:

1. *Document accuracy* refers to the proper coverage of your topics in appropriate detail. Often an accurate document needs to focus clearly on a problem. Document accuracy is generally cultivated by a clear problem statement and by a preliminary outline. These writing tools help you focus your writing effort by reducing your data in a way that solves a theoretical or practical problem.
2. *Stylistic accuracy* concerns the careful use of language to express meaning. Accurate language requires the careful use of paragraph and sentence structure and word choice to describe and analyze your topics effectively. As a writer, you gain command of accuracy by studying the elements of style and by learning to apply those elements to your drafting, revising, editing, and proofreading. Stylistic accuracy is also a matter of using words precisely.
3. *Technical accuracy* requires stylistic accuracy but is not based solely on it. The effective document in science and technology must be grounded in a technically accurate understanding and representation of the subject. Technical accuracy depends on the writer's conceptual mastery of the subject and its vocabulary, as well as on his or her ability to analyze and shape data with a minimum of distortion. In science and technology, enormous creative energy is given to mastering this technical aspect of subject development.

Section 1.1.2

Clarity

Strive for clarity in your writing. Clarity, which refers to ease of understanding, is a special problem in science and technology writing. Specialized languages, mathematically detailed analyses, and complex conceptual schemes can make technical subjects hard to grasp even when prepared by skilled writers and read by expert readers. You can increase the clarity of your material in several ways.

At the level of the whole document, you can promote structural clarity, making it easy for the reader to get the large picture. Use abstracts and other forecasting strategies such as introductions that state the purpose and scope of the document. Tables of contents, problem statements, and even strategic repetition also promote structural clarity. Graphs and tables, effectively designed and placed, help focus and clarify information. Descriptive titles and frequent subject headings guide readers and help keep the large picture in focus.

Stylistic clarity is promoted by simple, direct language. Simplicity in language is obtained with directly worded sentences. Using simple sentences and avoiding overloaded sentences and excessive nominalization also contribute to clarity. Word choice is a factor in stylistic clarity: use

simple language wherever possible to counteract the abstract, highly specialized terms of science and technology.

Contextual clarity, in which the importance, authorization, and implications of your work are made available, also contributes to ease of understanding. All work has a context, and your readers want to understand what the context of your document is. What prompts you to write? What is your purpose? Whose work precedes or has influenced yours? What is the organizational and intellectual context of your problem? You answer those questions in introductions and problem statements and in your citations and other references.

Section 1.1.3

Conciseness

Learn the strategies of conciseness. Conciseness has a special value in technical fields. Writers are often tempted to include everything that could be relevant to their subject, rather than merely everything that is relevant to the communication task at hand. The concise document is a piece of writing that conveys only the needed material.

At the level of the whole document, conciseness is helped most by *focus*, the narrowing of document scope to a manageable problem and response. Preparing a clear introduction and developing a detailed outline are two strategies that give you control over document length and scope. Identify and eliminate material that is not necessary to support your claims. Look for sections, including appendixes, that are not essential to your work. Graphics are powerful aids to conciseness because they cut down on the amount of prose necessary to describe objects and processes, summarize data, and demonstrate relationships.

Conciseness requires careful revising. Become familiar with the strategies for reducing wordiness. Look for ways of cutting useless words, sentences, and sections from the document.

Section 1.1.4

Coherence

Look for ways to improve the coherence of your writing. Coherence is the quality of hanging together, of providing the reader an easily followed path. Writers promote coherence by making their material logically and stylistically consistent, and by organizing and expressing their ideas in specific patterns. Efforts to emphasize the relationships among the elements of a document strengthen its impact. Coherence can dramatically improve the reader's ability to understand your material by promoting its flow or readability. Coherence is especially valued in science and technology because of the inherent complexity of the subjects.

At the level of the whole document, coherence helps to provide the larger picture, in which the connections among the parts of the document are made clear by the writer. Give readers a roadmap to help them anticipate the content of your work. Abstracts, clear titles, introductions, and problem statements all promote coherence by linking various parts of a piece of writing.

The paragraph is one of the most powerful instruments of coherence. By organizing material into a topic sentence and supporting sentences, paragraphs pull together material and emphasize various forms of conceptual development. Paragraph development is achieved partly through the specific

strategies of exemplification, analysis, comparison and contrast, definition, enumeration, and description, all of which furnish distinct approaches to developing ideas. Transitional devices also operate at the paragraph level to provide links between sentences and between paragraphs.

Section 1.1.5

Appropriateness

Make your document appropriate to your goals in writing it, your audience's purpose in reading it, and the specific institutional contexts in which it is written and read.

Because a reader's knowledge or experience determines the level of comprehension of technical material, appropriateness is largely determined by your audience. For example, a fact expressed in a mathematical equation may not be effective in a report addressed to a managerial audience. See Document Density.

All technical writing should also be appropriate to the specific institutional context that motivated its creation. It should not only serve the writer's and the reader's purposes but also conform to the goals and conventions of the institution in which it exists. Institutional goals and conventions are sometimes clear and explicit. For example, in large companies, the specific goals of various documents, as well as the preferred form and style, are often described in company correspondence and style manuals.

Although the context is not always clearly delineated, it always can be worked out. Class work should be done within the context of the goals of the class as well as the specific assignment. Research reports should conform to the general goals and specific conventions of the scientific or technical community in which they are created.

Section 1.3

Problem Statement

If you are focusing on a problem, be sure to define and state it specifically enough that you can writea bout it. Avoid trying to investigate or write about multiple problems or about broad or overly ambitious problems. Vague problem definition leads to unsuccessful proposals and vague, unmanageable documents. Naming a topic is not the same as defining a problem.

Weak

Coda file system. [This definition is too vague; it suggests a broad topic but not an approach to the topic.]

Improved

Protecting against temporary link failures in mobile computing: A design for the coda filesystem.

Weak

Engine starting and warm-up behavior.

Improved

Effects of fuel enrichment on engine starting and warm-up behavior.

Problem statements often have three elements:

1. the problem itself, stated clearly and with enough contextual detail to establish why it is important;
2. the method of solving the problem, often stated as a claim or a working thesis;
3. the purpose, statement of objective and scope of the document the writer is preparing.

These elements should be brief so that the reader does not get lost.

[problem and its context] A recent trend in the design of new aircraft is the addition of winglets, which are small fins attached to the ends of the main wing. After an aircraft has taken off and is cruising, winglets improve its performance by reducing the drag caused by the main wing. However, during the critical stages of aircraft takeoff and landing, the winglets cause two problems. First, they cause vibrations in the main wing, commonly called buffeting. Second, they cause the aircraft to lose some control of yaw, the motion of the nose right and left. In a study funded by NASA [Ref. 2], the main wing of a DC-10 transport aircraft was outfitted with winglets, and it experienced significant buffeting during takeoff and landing.

[approach of the current research] In our current project, we examine winglet-induced buffeting in three wing designs. We record buffeting and yaw under experimental wind-tunnel takeoff and landing conditions for (1) a wing without winglets, (2) another wing with conventional winglets, and (3) a wing with spheroid winglets. Our objective is to determine the degree to which differences between load lifts on the wings and their winglets during takeoff and landing are causing the performance problems we have described.

[purpose and scope of current document] In this study, we develop theoretical models of winglet load lifts and compare these to the lifts of wings and winglets actually recorded during testing conditions.

--Tan T. Trinh, "Winglets at Takeoffs and Landings"

Section 2.6

Proposals

In a proposal, identify a specific problem and state how you will solve that problem.

Most organizations rely on successful proposal writing for their continued existence. You will most likely spend a major part of your professional life writing proposals. Proposals are carefully prepared and just as carefully reviewed by granting agencies. Proposals do not succeed on the strength of a name or as a result of flashy rhetoric. Rather, successful proposals demonstrate that you understand the scope of the problem (its background, theory, and application) and, furthermore, that you have developed a valid and well-focused approach for reaching proposed objectives.

All proposals develop a plan of action in response to a specific need or problem. Some proposals are external, written in response to a request for proposals or an invitation to bid that has been published by an external organization. Other proposals are internal, written in response to a need within your own organization. In either case, your proposals must show that you understand the nature of the problem and that you have a specific and well-developed plan for arriving at a solution. Most proposals share a general structure for identifying the motivating problem, the objectives, and the proposed course of action.

Section 2.6.1

General Structure

Learn the basic structure of a proposal.

Proposals share a general document architecture, which is usually modified to suit specific circumstances. The overall structure of a proposal can be broken down into four parts:

Front matter

Body

- Introduction
- Technical approach
- Management requirements
- Work plan

End matter, or management requirements

Front Matter

The front matter of a proposal includes the following components:

Letter of transmittal

Title page

Summary

Table of contents

List of figures and tables

Body

Introduction
In the introduction to a proposal, do the following:

- Identify the motivating need or problem. Develop the immediate context in which this problem has been highlighted. In external proposals be sure you are responding to the published need and motivation as identified by the outside agency.
- Focus your proposal by stating the key technical issues, as you perceive them, underlying the motivating problem.
- State explicitly, and with unbiased language, your objectives.

Technical Approach
Identify and explain fully the technical approach you are taking to reach your objectives.

- Ground your technical approach in a brief explication of necessary theoretical background: applicable theory, analytical models, previous work, and hypotheses.
- Present your method and design for reaching your stated objectives. You will most likely make reference to figures in this section.
- State how pertinent data will be acquired, and just as important, describe your data analysis methods.

Management Requirements

Explain clearly how you will manage the development of your proposal project. Proposal reviewers pay strict attention to this section because here you show that you have the required know-how to bring a project to completion. In this section, you will present the following information:

A timetable (or Gantt chart)

Required facilities and resources

Materials and equipment

Personnel (include résumés in appendix)

Work Plan
Include a work plan, sometimes called a project plan, as separate section in all lengthy proposals. Preliminary project plans are also sometimes appropriate in feasibility and recommendation reports. In addition, most progress reports refer to all or part of previously existing project plans.

See Work Plan for a discussion of work plans.

End Matter

Bibliography

Résumés

Appendixes

Kinds of Proposals

Proposals may be written or oral, for government agencies or the private sector. Following is an example of a written government proposal. For an example of an oral presentation, Oral Presentations.

Section 3.4.7

Work Plan

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A project plan outlines in specific detail how a project will be conducted, who will work on which part, and when and in what order each part will be accomplished. It is similar to a procedure section but differs in that it describes future rather than past actions. Most project plans include the following elements:

- A short description of the project's objective
- A list of personnel participating in the project
- A list of all equipment and facilities to be used in the project
- A breakdown of the project into specific tasks, with indications of which tasks are dependent upon the completion of others
- A schedule indicating when each task will be started and when it will be completed and who will perform it; this information may be represented as an annotated bar chart
- A budget